Lecture 15

Formal characters and the Weyl denominator

Objectives. *The formal character of a module collects the dimensions of its weight spaces. We prove the rules for characters and show that a character determines a finite-dimensional module. We then compute the character of a Verma module and prove the Weyl denominator identity. These are the inputs of the character formula in Lecture 16.*

By the end of this lecture you will be able to:

- Compute characters of sums, tensor products and duals, and check Weyl invariance.
- Compute Verma weight multiplicities with Kostant's partition function.
- Prove the Weyl denominator identity and expand it for A_2.
- Decide when a difference of characters is the character of a module.

Throughout, $\mathfrak{g}$ is a nonzero complex semisimple Lie algebra with Cartan subalgebra $\mathfrak{h}$, positive roots R_+, simple roots $\alpha_1, \dots, \alpha_r$, coroots $\alpha_i^\vee = h_i$ and Weyl group W. Write $\langle \mu, \alpha^\vee \rangle = \mu(h_\alpha)$. Recall from Lecture 9 the root lattice $Q = \sum_i \mathbb{Z}\alpha_i$, the weight lattice P, the fundamental weights ω_i with $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$, and the set $P_+ = \sum_i \mathbb{Z}_{\geq 0}\omega_i$ of dominant weights. The letter ρ denotes the Weyl vector, half the sum of the positive roots. Since ρ is reserved for it, we write a module action with a dot, as in $H \cdot v$. Put

$$Q_+ = \Big\{ \sum_{i=1}^{r} k_i \alpha_i : k_i \in \mathbb{Z}_{\geq 0} \Big\}, \qquad \operatorname{ht}\Big(\sum_i k_i \alpha_i \Big) = \sum_i k_i,$$

and write $\mu \leq \lambda$ when $\lambda - \mu \in Q_+$.

15.1 Weights and the group ring

By Lecture 7 and Lecture 12, every finite-dimensional module V is the direct sum of its weight spaces

$$V_\mu = \{ v \in V : H \cdot v = \mu(H) v \text{ for all } H \in \mathfrak{h} \},$$

and every weight μ lies in P.

Definition 15.1 (Group ring and formal character)**.** The *group ring* $\mathbb{Z}[P]$ is the free abelian group with basis symbols e^μ, $\mu \in P$, and multiplication

$$e^\mu e^\nu = e^{\mu+\nu}.$$

The Weyl group acts on it by ring automorphisms $w(e^\mu) = e^{w\mu}$. The *formal character* of a finite-dimensional module V is

$$\operatorname{ch} V = \sum_{\mu \in P} \dim(V_\mu)\, e^\mu \in \mathbb{Z}[P].$$

By Lecture 9, W preserves P, so its action on $\mathbb{Z}[P]$ is well defined.
The symbol e^μ is not a function. Replacing every e^μ by 1 sends $\operatorname{ch} V$ to $\dim V$.

Example 15.2 ($\mathfrak{sl}_2$)**.** For $m \in \mathbb{Z}_{\geq 0}$, consider V_m. Here $P = \mathbb{Z}\omega$ with $\omega(h) = 1$, and $\alpha = 2\omega = 2\rho$. By Lecture 6, the basis vectors $f^k v_0$, $0 \leq k \leq m$, of V_m have weights $(m-2k)\omega$. Hence

$$\operatorname{ch} V_m = \sum_{k=0}^{m} e^{(m-2k)\omega} = e^{m\omega} + e^{(m-2)\omega} + \cdots + e^{-m\omega}.$$

15.2 Character rules and module isomorphism

Proposition 15.3 (Additivity, multiplicativity, duals · P)**.** *Let U, V and X be finite-dimensional modules.*

(a) If $0 \to U \to V \to X \to 0$ is exact, then $\operatorname{ch} V = \operatorname{ch} U + \operatorname{ch} X$.
(b) $\operatorname{ch}(V \otimes X) = \operatorname{ch} V \cdot \operatorname{ch} X$.
(c) $\operatorname{ch} V^* = \sum_\mu \dim(V_\mu)\, e^{-\mu}$.

Proof. (a) An intertwiner commutes with every $H \in \mathfrak{h}$. Hence it maps the μ-weight space into the μ-weight space. Each module is the direct sum of its weight spaces. Hence the sequence is the direct sum over μ of the sequences

$$0 \to U_\mu \to V_\mu \to X_\mu \to 0.$$

A direct sum of sequences is exact if and only if each summand is exact. Thus, for every μ,

$$\dim V_\mu = \dim U_\mu + \dim X_\mu.$$

This is the coefficient of e^μ in the claimed identity.

(b) Let $v \in V_\mu$ and $x \in X_\nu$. By the definition of the action on a tensor product,

$$H \cdot (v \otimes x) = (H \cdot v) \otimes x + v \otimes (H \cdot x) = (\mu + \nu)(H)\, v \otimes x.$$

Since $V \otimes X = \bigoplus_{\mu,\nu} V_\mu \otimes X_\nu$, we get

$$(V \otimes X)_\lambda = \bigoplus_{\mu+\nu=\lambda} V_\mu \otimes X_\nu, \qquad \dim(V \otimes X)_\lambda = \sum_{\mu+\nu=\lambda} \dim V_\mu \dim X_\nu.$$

The right-hand sum is the coefficient of e^λ in $\operatorname{ch} V \cdot \operatorname{ch} X$.

(c) Take a basis (v_k) of V made of weight vectors, with dual basis (v_k^*). Let $v_j \in V_\mu$ and $v_k \in V_{\mu_k}$. By the definition of the dual module,

$$(H \cdot v_j^*)(v_k) = -v_j^*(H \cdot v_k) = -\mu_k(H)\, \delta_{jk} = -\mu(H)\, v_j^*(v_k).$$

Hence $H \cdot v_j^* = -\mu(H) v_j^*$, and

$$\dim V^*_{-\mu} = \dim V_\mu.$$

∎

Theorem 15.4 (Weyl invariance · *P*). *For a finite-dimensional module V, every $w \in W$ and every $\mu \in P$,*

$$\dim V_{w\mu} = \dim V_\mu, \qquad \textit{equivalently} \qquad w(\operatorname{ch} V) = \operatorname{ch} V.$$

Proof. The simple reflections generate W (Lecture 9), so it suffices to treat s_i. Let (e_i, h_i, f_i) be the $\mathfrak{sl}_2$-triple of α_i (Lecture 8) and $\mathfrak{s}_i$ its span. Fix μ and put $m = \langle \mu, \alpha_i^\vee \rangle$. Consider

$$U = \bigoplus_{k \in \mathbb{Z}} V_{\mu + k\alpha_i}.$$

Since $e_i \in \mathfrak{g}_{\alpha_i}$ and $f_i \in \mathfrak{g}_{-\alpha_i}$, they map V_ν into $V_{\nu+\alpha_i}$ and $V_{\nu-\alpha_i}$, respectively. Thus U is an $\mathfrak{s}_i$-submodule. On $V_{\mu+k\alpha_i}$ the element h_i acts by

$$\langle \mu + k\alpha_i, \alpha_i^\vee \rangle = m + 2k.$$

Different k give different eigenvalues. So the $(m+2k)$-eigenspace of h_i on U is exactly $V_{\mu+k\alpha_i}$.

By complete reducibility (Lecture 5), U is a direct sum of modules V_n (Lecture 6). In V_n the eigenvalues $n, n-2, \dots, -n$ of h each occur once. Hence the eigenvalues m and $-m$ of h_i on U have equal multiplicity. The eigenvalue $-m$ occurs for $k = -m$, so

$$\dim V_\mu = \dim V_{\mu - m\alpha_i} = \dim V_{s_i\mu}.$$

∎

Proposition 15.5 (A character determines its module · *P*). *Finite-dimensional modules V and V' are isomorphic if and only if $\operatorname{ch} V = \operatorname{ch} V'$.*

Proof. An isomorphism $V \to V'$ maps each weight space V_μ isomorphically onto V'_μ. Hence isomorphic modules have equal characters.

For the reverse implication, we first show that the irreducible characters are linearly independent. By Lecture 12 and Lecture 13,

$$\operatorname{ch} L(\lambda) = e^\lambda + \sum_{\mu < \lambda} \dim L(\lambda)_\mu \, e^\mu \qquad (\lambda \in P_+).$$

Suppose a finite relation $\sum_\lambda c_\lambda \operatorname{ch} L(\lambda) = 0$ has some $c_\lambda \neq 0$. Choose λ maximal for $\leq$ among the weights with $c_\lambda \neq 0$. The term e^λ occurs in $\operatorname{ch} L(\lambda')$ only if $\lambda \leq \lambda'$. By maximality, among the λ' with $c_{\lambda'} \neq 0$ this happens only for $\lambda' = \lambda$. So the coefficient of e^λ in the relation is c_λ, and the relation forces $c_\lambda = 0$, a contradiction.

Next, by complete reducibility and Lecture 13, $V \cong \bigoplus L(\lambda)^{n_\lambda}$ and $V' \cong \bigoplus L(\lambda)^{n'_\lambda}$. Additivity gives

$$\sum_{\lambda \in P_+} (n_\lambda - n'_\lambda) \operatorname{ch} L(\lambda) = \operatorname{ch} V - \operatorname{ch} V' = 0.$$

Hence $n_\lambda = n'_\lambda$ for all λ, and $V \cong V'$. ∎

15.3 The Verma character

Departure from the standing scope. The Verma module $M(\lambda)$ is infinite-dimensional. Its character is an infinite sum, and we need a ring in which such sums can be multiplied.

Definition 15.6 (Completion). Let $\mathcal{E}$ be the set of formal sums $f = \sum_{\mu \in P} f_\mu e^\mu$ with $f_\mu \in \mathbb{Z}$, for which there are $\lambda_1, \dots, \lambda_s \in P$ with

$$\{\mu : f_\mu \neq 0\} \subseteq \bigcup_{j=1}^{s} (\lambda_j - Q_+).$$

A set $\lambda - Q_+$ is called the *cone* below λ. Add coefficientwise and multiply by

$$(fg)_\nu = \sum_{\mu + \mu' = \nu} f_\mu g_{\mu'}.$$

Lemma 15.7 (The completion is a ring · *P*). *Each product coefficient is a finite sum, and $fg \in \mathcal{E}$. Thus $\mathcal{E}$ is a commutative ring containing $\mathbb{Z}[P]$. For $\alpha \in R_+$,*

$$(1 - e^{-\alpha})^{-1} = \sum_{k \geq 0} e^{-k\alpha} \quad \text{in } \mathcal{E}.$$

Proof. Let the supports of f and g lie in $\bigcup_j (\lambda_j - Q_+)$ and $\bigcup_l (\lambda'_l - Q_+)$. If $f_\mu g_{\mu'} \neq 0$ and $\mu + \mu' = \nu$, then $\mu = \lambda_j - q$ and $\mu' = \lambda'_l - q'$ with $q, q' \in Q_+$, and

$$q + q' = \lambda_j + \lambda'_l - \nu.$$

For fixed j and l, the right side is fixed. An element of Q_+ has finitely many decompositions $q + q'$ in Q_+, because the simple-root coordinates of q lie between 0 and those of the sum. There are finitely many pairs (j, l), so the sum is finite, and the support of fg lies in $\bigcup_{j,l} (\lambda_j + \lambda'_l - Q_+)$.

Let $u \in \mathcal{E}$. The same argument with three cones shows that the coefficient of e^ν in $(fg)u$ and in $f(gu)$ is the same finite sum

$$\sum_{\mu + \mu' + \mu'' = \nu} f_\mu g_{\mu'} u_{\mu''}.$$

Hence the product is associative. Commutativity and distributivity follow directly from the formula for $(fg)_\nu$. Finally, the sum telescopes:

$$(1 - e^{-\alpha}) \sum_{k \geq 0} e^{-k\alpha} = \sum_{k \geq 0} e^{-k\alpha} - \sum_{k \geq 1} e^{-k\alpha} = 1. \qquad \blacksquare$$

Definition 15.8 (Character of a weight module). Let M be a $\mathfrak{g}$-module, possibly infinite-dimensional, such that $M = \bigoplus_{\mu \in P} M_\mu$, each M_μ is finite-dimensional, and every weight lies in a finite union $\bigcup_{j=1}^{s} (\lambda_j - Q_+)$. The *formal character* of M is

$$\operatorname{ch} M = \sum_{\mu \in P} \dim(M_\mu)\, e^\mu \in \mathcal{E}.$$

For a finite-dimensional module this is the earlier definition. The proof of Theorem 15.10 checks these hypotheses for $M(\lambda)$.

Definition 15.9 (Kostant partition function). Write p for $\mathcal{P}$ of Lecture 12. List $R_+ = \{\beta_1, \dots, \beta_N\}$. For $\gamma \in Q$ let

$$p(\gamma) = \#\Big\{(k_1, \dots, k_N) \in \mathbb{Z}_{\geq 0}^N : \sum_{j=1}^{N} k_j \beta_j = \gamma\Big\}.$$

Each β_j lies in Q_+ and has height at least 1. Hence $p(\gamma) = 0$ unless $\gamma \in Q_+$, and $p(0) = 1$, from the zero tuple alone. A solution therefore satisfies

$$k_j \le \sum_{l=1}^{N} k_l \,\mathrm{ht}\,\beta_l = \mathrm{ht}\,\gamma \qquad (1 \le j \le N).$$

Hence $p(\gamma)$ is finite. Expanding N geometric series gives

$$\prod_{\alpha\in R_+} \sum_{k\ge 0} e^{-k\alpha} = \sum_{(k_j)\in\mathbb{Z}_{\ge 0}^{N}} e^{-\sum_j k_j\beta_j} = \sum_{\gamma\in Q_+} p(\gamma) e^{-\gamma}. \tag{15.1}$$

Theorem 15.10 (Verma character · *P*). *Let $\lambda \in P$. The weights of $M(\lambda)$ lie in $\lambda - Q_+$, each weight space is finite-dimensional, and in $\mathcal{E}$*

$$\mathrm{ch}\, M(\lambda) = \sum_{\gamma\in Q_+} p(\gamma)\, e^{\lambda-\gamma} = e^{\lambda} \prod_{\alpha\in R_+} (1 - e^{-\alpha})^{-1}.$$

Proof. Choose nonzero $f_j \in \mathfrak{g}_{-\beta_j}$. By PBW (Lecture 11) and Lecture 12, the vectors

$$f^{\mathbf{k}} v_\lambda = f_1^{k_1} \cdots f_N^{k_N} v_\lambda, \qquad \mathbf{k} \in \mathbb{Z}_{\ge 0}^{N},$$

form a basis of $M(\lambda)$. Let $H \in \mathfrak{h}$. Since $f_j \in \mathfrak{g}_{-\beta_j}$, we have $[H, f_j] = -\beta_j(H) f_j$. Hence, in $U(\mathfrak{g})$,

$$H f_j = f_j H + [H, f_j] = f_j\big(H - \beta_j(H)\big).$$

Moving H to the right through each factor, and using $H \cdot v_\lambda = \lambda(H) v_\lambda$, gives

$$H \cdot f^{\mathbf{k}} v_\lambda = f^{\mathbf{k}}\Big(H - \sum_j k_j \beta_j(H)\Big) v_\lambda = \Big(\lambda - \sum_j k_j \beta_j\Big)(H)\, f^{\mathbf{k}} v_\lambda.$$

Hence each basis vector is a weight vector, and its weight lies in $\lambda - Q_+$. Grouping the basis by weight, $M(\lambda)_{\lambda-\gamma}$ has basis the $f^{\mathbf{k}} v_\lambda$ with $\sum_j k_j\beta_j = \gamma$. Its dimension is the finite number $p(\gamma)$. Equation (15.1) and Lemma 15.7 give the product form. ∎

15.4 Alternants and the Weyl denominator

Let $\det(w)$ be the determinant of w on $\mathfrak{h}^*$. A reflection negates a line and fixes a hyperplane, so $\det(s_\alpha) = -1$. Since W is generated by reflections, $\det(w) = \pm 1$.

Definition 15.11 (Alternant). For $\nu \in P$, the *alternant* is

$$A_\nu = \sum_{w\in W} \det(w)\, e^{w\nu} \in \mathbb{Z}[P].$$

Let $u \in W$. As w runs over W, so does $w' = uw$, and $\det(w) = \det(u)\det(w')$ because $\det(u) = \pm 1$. Hence

$$u A_\nu = \sum_{w} \det(w) e^{uw\nu} = \det(u) \sum_{w'} \det(w') e^{w'\nu} = \det(u) A_\nu.$$

An element $f \in \mathbb{Z}[P]$ with $wf = \det(w) f$ for all $w \in W$ is called *anti-invariant*. So every alternant is anti-invariant.

Lemma 15.12 (The Weyl vector · *P*). *Each s_i permutes $R_+ \setminus \{\alpha_i\}$. Hence $s_i\rho = \rho - \alpha_i$ and $\langle \rho, \alpha_i^\vee \rangle = 1$.*

Proof. Let $\beta \in R_+$, $\beta \neq \alpha_i$, and write $\beta = \sum_j c_j \alpha_j$ with $c_j \geq 0$. Roots are reduced (Lecture 8), so $c_j > 0$ for some $j \neq i$. The root

$$s_i\beta = \beta - \langle \beta, \alpha_i^\vee \rangle \alpha_i$$

has the same coefficient $c_j > 0$. Every root is positive or negative (Lecture 9), so $s_i\beta$ is positive. Moreover, $s_i\beta \neq \alpha_i$, since $s_i\beta = \alpha_i$ would give $\beta = s_i\alpha_i = -\alpha_i \notin R_+$. Thus s_i maps the finite set $R_+ \setminus \{\alpha_i\}$ injectively, hence bijectively, into itself. Since also $s_i\alpha_i = -\alpha_i$,

$$s_i\rho = \frac{1}{2} \sum_{\beta \neq \alpha_i} \beta - \frac{1}{2}\alpha_i = \rho - \alpha_i.$$

Comparing with $s_i\rho = \rho - \langle \rho, \alpha_i^\vee \rangle \alpha_i$ gives the last claim. ∎

Hence $\rho = \omega_1 + \cdots + \omega_r \in P_+$. We import one fact about stabilisers from Lecture 9.

Lemma 15.13 (Stabilisers of dominant weights · *C*). *Let $\lambda \in P$ be dominant and $w \in W$ with $w\lambda = \lambda$. Then w is a product of simple reflections fixing λ. In particular, if $\langle \lambda, \alpha_i^\vee \rangle > 0$ for all i, then $w = 1$. Source: Lecture 9; Humphreys, Section 10.3, Lemma B, for a vector in the closed fundamental chamber of a root system with a fixed base.*

Theorem 15.14 (Weyl denominator identity · *P*). *In $\mathbb{Z}[P]$,*

$$A_\rho = e^\rho \prod_{\alpha \in R_+} (1 - e^{-\alpha}).$$

Proof. Write Δ for the right side. It lies in $\mathbb{Z}[P]$ because $\rho \in P$.

Step 1: anti-invariance. By Lemma 15.12,

$$s_i\Delta = e^{\rho - \alpha_i}(1 - e^{\alpha_i}) \prod_{\alpha \in R_+,\ \alpha \neq \alpha_i} (1 - e^{-\alpha}),$$

and

$$e^{-\alpha_i}(1 - e^{\alpha_i}) = e^{-\alpha_i} - 1 = -(1 - e^{-\alpha_i}).$$

Hence $s_i\Delta = -\Delta$. The s_i generate W and $\det(s_i) = -1$, so $w\Delta = \det(w)\Delta$. In other words, Δ is anti-invariant.

Step 2: support. Expanding over subsets $S \subseteq R_+$,

$$\Delta = \sum_{S \subseteq R_+} (-1)^{|S|}\, e^{\rho - \langle S \rangle}, \qquad \langle S \rangle = \sum_{\alpha \in S} \alpha.$$

Write $\Delta = \sum_\mu c_\mu e^\mu$. Each $\langle S \rangle$ lies in Q_+, so every μ with $c_\mu \neq 0$ satisfies $\mu \leq \rho$. A nonempty S has $\mathrm{ht}\langle S \rangle > 0$, so $c_\rho = 1$.

Step 3: dominant terms. Let $w \in W$. The coefficient of $e^{w\mu}$ is c_μ in $w\Delta$ and $\det(w)c_{w\mu}$ in $\det(w)\Delta$. Since $\det(w) = \pm 1$, this gives

$$c_{w\mu} = \det(w)\, c_\mu \qquad (w \in W,\ \mu \in P).$$

Let $c_\mu \neq 0$ and let ν be the dominant element of $W\mu$ (Lecture 9). Then $c_\nu \neq 0$. If $\langle \nu, \alpha_i^\vee \rangle = 0$, then $s_i\nu = \nu$ and $c_\nu = -c_\nu = 0$. Hence $\langle \nu, \alpha_i^\vee \rangle \geq 1$ for all i, so $\lambda = \nu - \rho \in P_+$.

Step 4: only ρ. Step 2 gives $\nu \le \rho$. So $-\lambda = \sum_i k_i \alpha_i$ with $k_i \ge 0$. Take the positive definite W-invariant form of Lecture 8. Then

$$\begin{aligned}(\lambda,\lambda) &= -\sum_i k_i(\lambda,\alpha_i)\\ &= -\sum_i k_i \frac{1}{2}(\alpha_i,\alpha_i)\langle\lambda,\alpha_i^\vee\rangle\\ &\le 0.\end{aligned}$$

The second line uses $\langle\lambda,\alpha_i^\vee\rangle = 2(\lambda,\alpha_i)/(\alpha_i,\alpha_i)$. Positive definiteness gives $\lambda = 0$, so $\nu = \rho$. Hence all nonzero coefficients lie on $W\rho$.

Step 5: the orbit. By Lemmas 15.12 and 15.13, $w \mapsto w\rho$ is a bijection $W \to W\rho$. So

$$\Delta = \sum_{w\in W} c_{w\rho} e^{w\rho} = \sum_{w\in W} \det(w) c_\rho e^{w\rho} = A_\rho. \qquad \blacksquare$$

The proof uses no character formula. Combined with Theorem 15.10, it gives the following.

Corollary 15.15 (Verma characters over the denominator · *P*)**.** *A_ρ is invertible in $\mathcal{E}$, and $\operatorname{ch} M(\lambda)\cdot A_\rho = e^{\lambda+\rho}$ for every $\lambda \in P$.*

Proof. By Lemma 15.7, each factor $1 - e^{-\alpha}$ is invertible in $\mathcal{E}$. Theorems 15.10 and 15.14 give

$$\operatorname{ch} M(\lambda)\cdot A_\rho = e^\lambda \prod_{\alpha\in R_+}(1-e^{-\alpha})^{-1}\cdot e^\rho \prod_{\alpha\in R_+}(1-e^{-\alpha}) = e^{\lambda+\rho}.$$

For $\lambda = -\rho$ the right side is $e^0 = 1$. Thus $\operatorname{ch} M(-\rho)$ is the inverse of A_ρ. $\blacksquare$

Consequently, for $\lambda \in P$, the quotient $A_{\lambda+\rho}/A_\rho$ is defined in $\mathcal{E}$. For $w \in W$, apply the corollary at the weight $w(\lambda+\rho)-\rho$ and multiply by A_ρ^{-1}. This gives

$$\operatorname{ch} M\big(w(\lambda+\rho)-\rho\big) = e^{w(\lambda+\rho)} A_\rho^{-1}.$$

Hence

$$\begin{aligned}\frac{A_{\lambda+\rho}}{A_\rho} &= \sum_{w\in W} \det(w)\, e^{w(\lambda+\rho)}\cdot A_\rho^{-1}\\ &= \sum_{w\in W} \det(w)\operatorname{ch} M\big(w(\lambda+\rho)-\rho\big).\end{aligned}$$

Lecture 16 proves that this element is $\operatorname{ch} L(\lambda)$ for $\lambda \in P_+$.

15.5 The A_2 calculation

Example 15.16 (The A_2 denominator)**.** For $\mathfrak{sl}_3$, $R_+ = \{\alpha_1,\alpha_2,\theta\}$ with $\theta = \alpha_1+\alpha_2$, so $\rho = \alpha_1+\alpha_2$. Put $x = e^{-\alpha_1}$ and $y = e^{-\alpha_2}$. Then

$$\begin{aligned}(1-x)(1-y) &= 1 - x - y + xy,\\ (1-x-y+xy)(1-xy) &= 1 - x - y + xy - xy + x^2y + xy^2 - x^2y^2\\ &= 1 - x - y + x^2y + xy^2 - x^2y^2.\end{aligned}$$

Two of the eight subset terms cancel. They come from $S = \{\alpha_1,\alpha_2\}$ and $S = \{\theta\}$, which both give the weight $\rho - \theta = 0$, with opposite signs. Every reflection fixes this weight.

Now compute the orbit $W\rho$ under the six elements $1, s_1, s_2, s_1s_2, s_2s_1, s_1s_2s_1$ of W, using $\langle\alpha_1, \alpha_2^\vee\rangle = \langle\alpha_2, \alpha_1^\vee\rangle = -1$:

$$\begin{aligned} s_1\rho &= \rho - \alpha_1, \qquad s_2\rho = \rho - \alpha_2, \\ s_1s_2\rho &= s_1(\rho - \alpha_2) = \rho - \alpha_1 - (\alpha_1 + \alpha_2) = \rho - 2\alpha_1 - \alpha_2, \\ s_2s_1\rho &= \rho - \alpha_1 - 2\alpha_2, \qquad s_1s_2s_1\rho = -\rho = \rho - 2\alpha_1 - 2\alpha_2. \end{aligned}$$

A product of k simple reflections has determinant $(-1)^k$. So the six signs, in this order, are $+, -, -, +, +, -$. Hence

$$A_\rho = e^\rho - e^{\rho-\alpha_1} - e^{\rho-\alpha_2} + e^{\rho-2\alpha_1-\alpha_2} + e^{\rho-\alpha_1-2\alpha_2} - e^{-\rho},$$

which is $e^\rho(1 - x - y + x^2y + xy^2 - x^2y^2)$, as the theorem asserts.

Example 15.17 (Verma multiplicities for A_2)**.** Take $f_1 = E_{21}$, $f_2 = E_{32}$, $f_3 = E_{31}$, in the root spaces of $-\alpha_1, -\alpha_2, -\theta$. For $\gamma = a\alpha_1 + b\alpha_2$ with $a, b \in \mathbb{Z}_{\geq 0}$, the equation $k_1\alpha_1 + k_2\alpha_2 + k_3\theta = \gamma$ means

$$k_1 + k_3 = a, \qquad k_2 + k_3 = b.$$

Each k_3 with $0 \leq k_3 \leq \min(a, b)$ gives the solution $k_1 = a - k_3$, $k_2 = b - k_3$, and these are all the solutions. Hence

$$p(a\alpha_1 + b\alpha_2) = \min(a, b) + 1.$$

Exercise 15.3 gives PBW bases and coefficient checks.

Subtracting characters

Characters are elements of the abelian group $\mathcal{E}$. An identity there is an identity of integer-valued functions on P. Subtraction is defined in $\mathcal{E}$, but a difference of characters need not be the character of a module.

For an exact sequence $0 \to U \to V \to X \to 0$ of finite-dimensional modules, additivity gives $\operatorname{ch} V - \operatorname{ch} U = \operatorname{ch} X$. More generally, let f be a difference of characters of finite-dimensional modules. By complete reducibility, f is an integer combination of the characters $\operatorname{ch} L(\lambda)$, $\lambda \in P_+$. The coefficients are unique, by the linear independence shown in the proof of Proposition 15.5. Hence f is the character of a finite-dimensional module exactly when all these coefficients are nonnegative. Nonnegative W-invariant weight coefficients do not suffice. Exercise 15.1 gives an example.

Self-checks

(1) For A_2, what is $\dim M(\lambda)_{\lambda-3\alpha_1-\alpha_2}$?
(2) Does $\sum_{k\geq 0} e^{k\alpha}$, with $\alpha \in R_+$, lie in $\mathcal{E}$?
(3) Why must the coefficient of e^0 in $e^\rho(1-x)(1-y)(1-xy)$ vanish without expanding?

Answers. (1) $\min(3, 1) + 1 = 2$, with basis $f_1^3 f_2 v_\lambda$ and $f_1^2 f_3 v_\lambda$. (2) No. A cone $\lambda_j - Q_+$ contains $k\alpha$ only if the simple-root coordinates of $\lambda_j - k\alpha$ are nonnegative, which fails for large k. So each of the finitely many cones contains only finitely many $k\alpha$, and their union misses the rest. (3) The product is anti-invariant and s_1 fixes 0, so $c_0 = -c_0$.

Summary

For a finite-dimensional module, the formal character records the dimensions of its weight spaces. It is additive, multiplicative and Weyl-invariant. It determines the module up to isomorphism. A difference of finite-dimensional characters is the character of a module exactly when its coefficients in the irreducible characters are nonnegative. The PBW basis gives $\operatorname{ch} M(\lambda) = \sum_\gamma p(\gamma) e^{\lambda-\gamma}$ for $\lambda \in P$ in the completion $\mathcal{E}$, whose elements are supported in finitely many cones.

The Weyl denominator identity follows from anti-invariance, the support bound and positivity of the form. Together with the Verma character, it shows that A_ρ is the inverse of $\operatorname{ch} M(-\rho)$.

Reading

J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 22–24 and Section 10.3. B. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Sections 10.4 and 10.6–10.7.